SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO4 AT1 - Big Data Analytics Class Test Student Template

Course Code	CSA15	**Slot**	B
Course Title	Cloud Computing and Big Data Analytics	**Assessment**	CO4 AT1
Assessment Title	Big Data Analytics Class Test	**Bloom Level**	BL4 - Analyze
Faculty	Dr C. Rajesh Babu	**Employee ID**	SSETSCS415
Submission Mode	Individual digital answer template through GitHub and Google Classroom	**Total Marks**	30

Student Details

Student Name	Karanam Reddy Vishnu Chaithanya	**Register Number**	192372057
Class / Section		**Slot**	B
Capstone Title	Sentinel X Cloud Threat Intelligence Using NLP and Serverless Risk Scoring	**Submission Date**	
GitHub Repository Link		**Google Classroom Status**	

Assessment Overview

CO4 Focus

Analyze big data characteristics, processing needs, security challenges, business drivers and domain applications. This template includes explanation notes to guide the thinking process. The final answer must be written in the student's own words.

How to use this template:

- Read each question and the explanation notes before answering.
- Use your capstone product as an example wherever relevant.
- Write concise technical answers; avoid copying the explanation notes directly.
- Use diagrams or small tables wherever they improve clarity.
- Upload the completed DOCX/PDF and supporting evidence to GitHub; share the repository link through Google Classroom.

Assessment Rubric

Criteria	Marks	Evaluation Focus	Student Check

Concept Accuracy	9	Correct technical meaning, terminology and examples.	Have I used correct big data terms?
Coverage	6	All required parts of the question are answered.	Have I answered every part?
Examples	4.5	Relevant cloud/capstone/domain examples are included.	Have I used meaningful examples?
Analysis	6	Implications, security, scalability and decision value are explained.	Have I explained why it matters?
Clarity	4.5	Answer is organized, concise and readable.	Is my answer easy to evaluate?

Code of Conduct and Student Assurance

I Karanam Reddy Vishnu Chaithanya /192372057 (Name / Register Number) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.R.V. CHAITHANYA

Question and Answer Sections

Answer all questions. Each question carries 3 marks. Use the explanation notes for direction, but write the final answer independently.

Question 1 (3 Marks)

Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.

Explanation Notes

1	Start with a precise definition: Big Data refers to datasets whose size, speed, variety or complexity exceeds the capability of conventional tools.
2	Mention why spreadsheets or single-server databases struggle: storage limits, slow queries, concurrency issues, schema rigidity and limited scalability.
3	Use one cloud/capstone-style example such as app logs, IoT feeds, transactions, images or clickstream events.
4	End with why distributed storage and parallel processing become necessary.

Student Response

Big Data refers to very large and complex datasets whose size, speed, variety, or complexity exceeds the capabilities of conventional data-processing tools. Traditional spreadsheets and single-server databases may face storage limitations, slow queries, concurrency problems, rigid schemas, and limited scalability when handling such data.
For example, Sentinel X can handle large amounts of cybersecurity data such as threat reports, security logs, alerts, and text-based threat information. As the data grows continuously, a single server may not process it efficiently.
Therefore, distributed storage and parallel processing are required to store and analyse large datasets efficiently and support scalable Big Data applications

Question 2 (3 Marks)

Explain the 5Vs of Big Data with one practical example for any two Vs.

Explanation Notes

1	List the 5Vs clearly: Volume, Velocity, Variety, Veracity and Value.
2	For at least two Vs, connect the definition to a concrete scenario instead of writing only textbook meaning.
3	Example: In a smart traffic system, velocity refers to continuous vehicle movement updates; value refers to congestion prediction.
4	Show that each V affects system design, storage, processing or analytics decisions.

Student Response

The 5Vs of Big Data are:

1. Volume – The huge amount of data generated and stored.
2. Velocity – The speed at which data is generated and processed.
3. Variety – Different forms of data such as text, images, logs, and JSON.
4. Veracity – The accuracy, reliability, and quality of data.
5. Value – The useful information obtained from analysing the data.

In SentinelX, Volume can represent the large number of cybersecurity logs and threat records collected from different sources. Velocity represents continuously arriving security alerts that need quick processing.
These Vs influence the system design because they determine the required storage capacity, processing speed, data formats, quality controls, and analytics methods.

Question 3 (3 Marks)

Differentiate structured, semi-structured, and unstructured data with examples.

Explanation Notes

1	Structured data has a fixed schema, such as customer table, transaction table or attendance record.
2	Semi-structured data has tags/keys but flexible structure, such as JSON, XML, API responses or log files.
3	Unstructured data has no fixed tabular form, such as images, videos, audio, PDFs, emails or social media text.
4	Use examples from one cloud-backed application to make the comparison coherent.

Student Response

Data Type	Meaning	Example in SentinelX
Structured	Data stored in a fixed schema such as rows and columns.	Threat ID, severity, date, and status stored in a database table
Semi-structured	Data does not follow a fixed table structure but contains keys, tags, or other organizational elements.	JSON-formatted security API responses or application logs
Unstructured	Data has no fixed tabular structure.	Threat reports, security documents, text descriptions, images, or PDFs

SentinelX may need to process all three types because cybersecurity information can come from databases, APIs/logs, and text-based threat intelligence sources. Therefore, the system should support flexible storage and processing methods for different data formats

Question 4 (3 Marks)

Explain the role of distributed file systems in big data processing.

Explanation Notes

1	State the main need: very large datasets are split and stored across multiple machines.
2	Mention core benefits: scalability, fault tolerance, parallel access and support for batch processing.
3	Use HDFS or cloud object storage as an example, but do not confuse file storage with relational database storage.
4	Connect storage to processing: data blocks can be processed in parallel by frameworks such as MapReduce or Spark.

Student Response

A distributed file system stores very large datasets by splitting the data into blocks and distributing them across multiple machines. This provides scalability, fault tolerance, parallel access, and efficient batch processing.
For example, HDFS is a distributed file system, while cloud object storage can also be used for large-scale data storage.
In a SentinelX-type system, large collections of threat intelligence data can be distributed across cloud storage instead of depending on one machine.

The stored data blocks can then be processed in parallel using frameworks such as MapReduce or Spark, making Big Data analysis faster and more scalable.

Question 5 (3 Marks)

List any three major security concerns in big data platforms and briefly explain each.

Explanation Notes

1	Choose three concerns such as unauthorized access, data leakage, privacy violation, weak encryption, poor audit logs or insecure APIs.
2	For each concern, explain the risk and not only the keyword.
3	Add one control where possible: role-based access, encryption, anonymization, masking, logging or monitoring.
4	Show awareness that big data combines many data sources, so security and governance become more complex.

Student Response

Three major security concerns are:

1. Unauthorized Access: Attackers or unauthorized users may access sensitive Big Data. Role-Based Access Control (RBAC) can restrict users according to their permissions.
2. Data Leakage: Sensitive information may be exposed during storage or transmission. Encryption can protect data from unauthorized access.
3. Privacy Violations: Combining data from multiple sources can reveal sensitive information about individuals or organizations. Data masking, anonymization, and proper access controls can reduce this risk.

For Sentinel X, security is especially important because threat intelligence may contain sensitive security information. Therefore, access control, encryption, logging, and monitoring should be included in the system.

Question 6 (3 Marks)

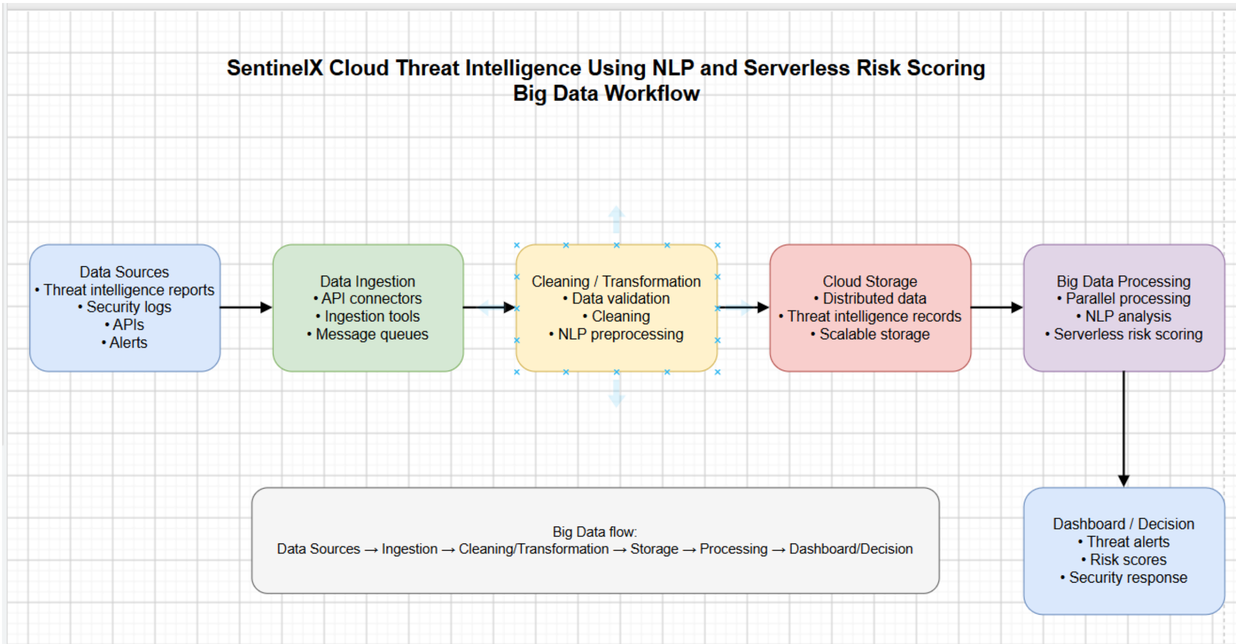
What are data appliances and integration tools? Explain their role in big data workflows.

Explanation Notes

1	A data appliance is a specialized hardware/software platform optimized for storage, processing or analytics workloads.
2	Integration tools move, transform or synchronize data between sources, storage systems and analytics layers.
3	Mention ETL/ELT, API connectors, message queues, ingestion tools or workflow schedulers as examples.
4	Explain the workflow: source data -> ingestion -> cleaning/transformation -> storage -> processing -> dashboard/decision.

Student Response

A data appliance is a specialized hardware or software platform designed and optimized for data storage, processing, or analytics. Integration tools are used to move, transform, and synchronize data between different sources, storage systems, and analytics layers.



In Sentinel X, threat intelligence from different sources can be collected through APIs or ingestion mechanisms, cleaned and transformed, stored in cloud infrastructure, and then analyzed to produce useful threat information and risk scores.

Question 7 (3 Marks)

Discuss two major challenges organizations face while implementing big data analytics.

Explanation Notes

1	Pick two strong challenges such as data quality, skill gap, infrastructure cost, privacy compliance, integration difficulty or real-time scalability.
2	Explain the cause and impact of each challenge.
3	Suggest a practical mitigation for each challenge, such as governance, cloud scaling, training, access control or phased implementation.
4	Try to connect the challenge to business impact, not only technical difficulty.

Student Response

Two major challenges are:

1. Data Quality: Big Data can come from many sources and may contain incomplete, duplicate, outdated, or inaccurate information. Poor-quality data can produce unreliable analytical results. This can be reduced through data validation, cleaning, and governance.
2. Real-Time Scalability: Some applications receive data continuously and require quick analysis. As the data volume increases, the system must scale its processing capacity. Cloud-based scaling and distributed processing can help manage changing workloads.

For SentinelX, maintaining accurate threat intelligence and processing incoming security information quickly are important because incorrect or delayed analysis can affect threat detection and risk assessment.

Question 8 (3 Marks)

Write short notes on any one domain application of big data analytics.

Explanation Notes

1	Select one domain: healthcare, retail, agriculture, finance, education, smart city, transport or cybersecurity.
2	Identify data sources in that domain.
3	Explain how analytics converts raw data into decisions: prediction, recommendation, alert, optimization or risk detection.
4	Mention one expected outcome such as reduced waiting time, fraud detection, crop advisory or improved user experience.

Student Response

Big Data Analytics has an important application in cybersecurity. Data can be collected from security logs, threat reports, network events, alerts, APIs, and other security sources.
In SentinelX, analytics can process threat intelligence and textual security information to identify patterns, detect potential threats, and generate risk scores or alerts. NLP can help extract useful information from text-based threat intelligence.
The main outcome is improved threat detection, risk identification, faster response, and better cybersecurity decision making.

Question 9 (3 Marks)

How do business drivers influence the adoption of big data solutions?

Explanation Notes

1	Business drivers are reasons that push an organization to adopt big data, such as cost reduction, personalization, speed, risk control or better decisions.
2	Explain how each selected driver creates a technical requirement.
3	Example: personalization requires behavior tracking and recommendation analytics; risk control requires fraud detection and monitoring.
4	Conclude that big data adoption must be justified by measurable business value.

Student Response

Business drivers are the reasons that encourage organizations to adopt Big Data solutions. Important drivers include cost reduction, faster processing, personalization, risk control, and better decision making.
For example, risk control can create a requirement for continuous monitoring and threat detection. Similarly, the need for faster decisions requires systems capable of processing data quickly and producing useful insights.
For Sentinel X, the business driver is improved cybersecurity risk control. Threat intelligence analytics can help organizations identify risks earlier and make faster security decisions. Therefore, Big Data adoption should provide measurable business value, such as improved threat detection and reduced response time.

Question 10 (3 Marks)

Explain the importance of big data analytics in decision making.

Explanation Notes

1	Explain that analytics helps convert raw, high-volume data into useful patterns, predictions, alerts or recommendations.
2	Differentiate descriptive, diagnostic, predictive and prescriptive decision support if space permits.
3	Use one example: dashboard for hospital beds, forecast for demand, fraud alert for bank, traffic prediction for smart city.
4	End with the idea that decisions become faster, evidence-based and scalable.

Student Response

Big Data Analytics converts large amounts of raw data into useful patterns, predictions, alerts, and recommendations, helping organizations make faster and evidence-based decisions.
Decision support can be understood through four levels:

- Descriptive: What happened?
- Diagnostic: Why did it happen?
- Predictive: What may happen next?
- Prescriptive: What action should be taken?

In Sentinel X, analytics can examine threat intelligence and generate risk scores and alerts, helping security teams identify potentially dangerous threats and decide how to respond.
Thus, Big Data Analytics makes decision making faster, more accurate, evidence-based, and scalable.

Submission Checklist

Checklist Item	Status	Remarks
Student details completed.	Done	
All 10 questions answered.	Done	
Examples are connected to cloud/capstone/domain context.	Done	
Security/privacy points are included wherever relevant.	Done	
Document exported as PDF if requested.	Done	
GitHub link shared through Google Classroom.	Done	